



Paper ID : 259460

Printed Page: 1 of 2
Subject Code: RCAI401

Roll No:

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MCA-INT
(SEM IV) THEORY EXAMINATION 2024-25
COMPUTER BASED STATISTICAL TECHNIQUES

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.

02 x 7 = 14

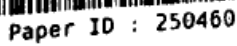
Q no.	Question	CO	Level
a.	Why do Floating point arithmetic operations sometimes results in rounding errors? Provide an example.	1	K1
b.	Given the number 1.234567 in single-precision floating-point representation, what is the closest approximation (with limited precision) to 1.23456789?	1	K1
c.	Write the Lagrange's Interpolation Formula.	2	K1
d.	Write the general formula for Newton's Backward Interpolation?	2	K1
e.	Given the function $f(x)=x^2$ over the interval $[1,3]$, and using the Trapezoidal Rule with $n=2$, what is the approximate value of the integral?	3	K2
f.	What is the importance of pivoting in the Gauss Elimination method?	4	K2
g.	State the common methods used for curve fitting.	5	K1

SECTION B

2. Attempt any three of the following:

07 x 3 = 21

Q no.	Question	CO	Level												
a.	Define averaging operator μ and central difference operator δ . Given the function $f(x) = e^x$, apply both the averaging operator μ and the central difference operator δ at $x=1$, where $h=0.5$, and compare their results.	1	K2												
b.	Given the following values of $f(x)$ <table border="1" style="margin: 5px auto;"> <tr> <td>x</td><td>0</td><td>2</td><td>4</td><td>6</td><td>8</td></tr> <tr> <td>f(x)</td><td>1</td><td>5</td><td>13</td><td>25</td><td>41</td></tr> </table> Use Bessel's formula to estimate the $f(5)$.	x	0	2	4	6	8	f(x)	1	5	13	25	41	2	K2
x	0	2	4	6	8										
f(x)	1	5	13	25	41										
c.	Use Simpson's Rule to estimate the integral of the function $f(x)=\sin(x)$ from $x=0$ to $x=\pi$ with 4 equal subintervals. $\int_0^\pi \sin(x) dx$	3	K3												
d.	Solve the system of linear equations using the Gauss-Seidel iterative method: $4x+y-z=3$ $x+3y+2z=12$ $2x+2y+3z=10$	4	K1												
e.	Explain Linear regression. Given the following data points, fit a straight line $y=mx+c$ using the method of least square. <table border="1" style="margin: 5px auto;"> <tr> <td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr> <tr> <td>y</td><td>2</td><td>3</td><td>5</td><td>7</td><td>8</td></tr> </table>	x	1	2	3	4	5	y	2	3	5	7	8	5	K3
x	1	2	3	4	5										
y	2	3	5	7	8										



Roll No:

TIME: 3 HRS

M.MARKS: 7

3. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

4. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

5. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

6. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

7. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

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